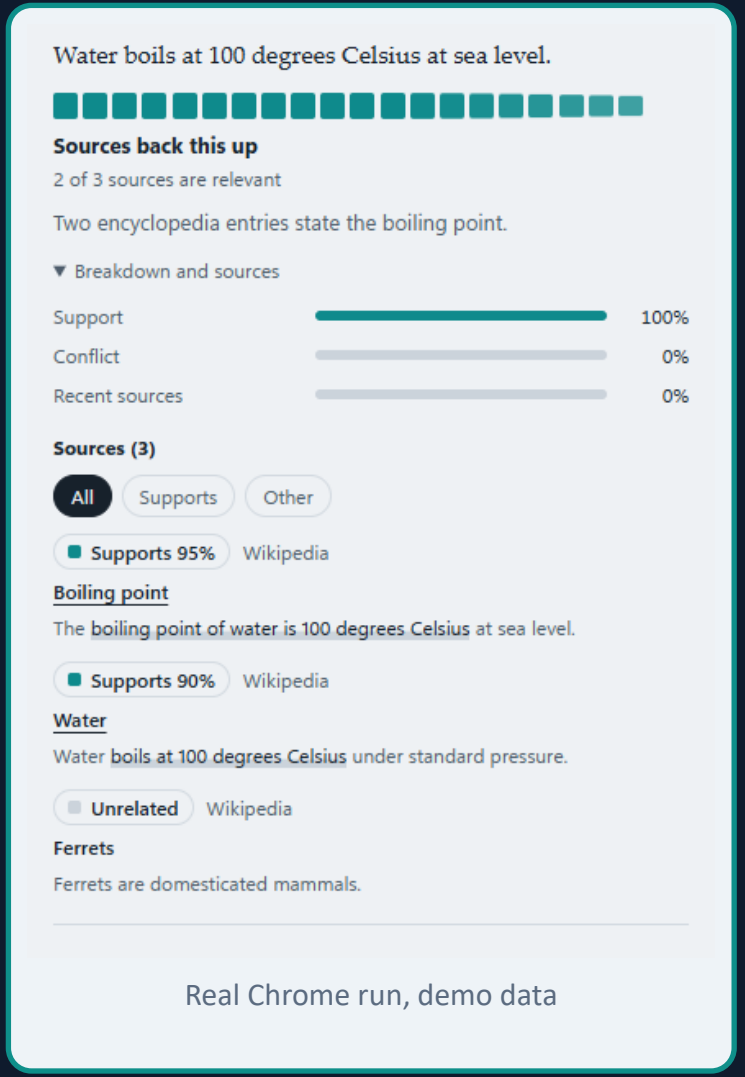
RESEARCH IMPLEMENTATION REVIEW 1

Synthetic Silence 2.0

Auditing language-model answers against sources: deterministic verdict rules, provable properties, and a validation on FEVER

supported | refuted | mixed | insufficient

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People read LLM answers with no way to check them



Fluent and specific

Names, dates and figures make an invented fact look like a recalled one.



No marker

A reader cannot tell recall from invention. Larger models can repeat common misconceptions more often (Lin et al., 2022).



Silence on failure

A failed search, a missing source and a disagreeing source must not look like the same outcome.

Problem statement

Given an answer written by a language model, return one verdict per claim that a reader can trace to a quoted sentence in a source, and keep “no evidence” apart from “the check failed”.

Five objectives, each tied to something we can test

1. **Verdict procedure**
   Per-source stances with a verbatim quote; five outcomes.

2. **Provable properties**
   Prove them for the rules and two earlier scores; check against the code.

3. **Validation on FEVER**
   Confusion matrices, Wilson intervals, paired exact tests.

4. **Explicit failure**
   An outage is an error for that claim, never “no sources”.

5. **Working tool**
   Chrome extension on a FastAPI backend, plus a cross-model audit protocol.

Research questions

RQ1 Does the retrieval source or the aggregation rule matter more for decisive claims?

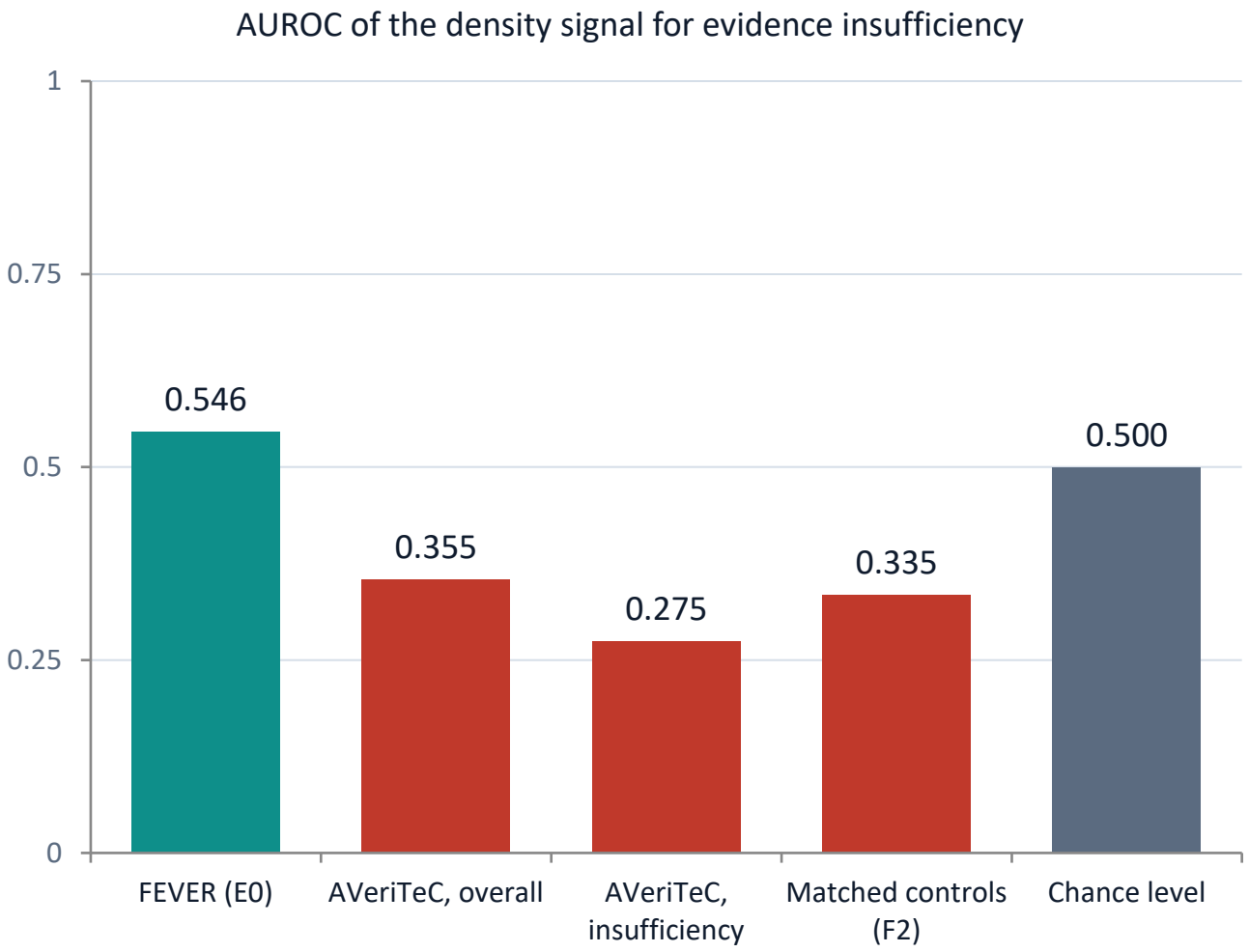
RQ2 Can a pooled entailment score abstain when sources do not mention a claim?

RQ3 Does a density or entropy score track whether evidence supports a claim?

Literature: no quote-backed audit at reading time

Work	What it does	Gap for a reader-side audit
Guo et al., 2022	Survey: claim detection, retrieval, verdict, justification	Pipeline view; no deployed audit
Thorne et al., 2018 (FEVER)	Wikipedia claims labelled SUPPORTS, REFUTES, NEI	Fixed corpus; claims given, not extracted
Schlichtkrull et al., 2023 (AVeriTeC)	Real-world claims with open-web evidence	A dataset, not a reader-facing verdict
Manakul et al., 2023 (SelfCheckGPT)	Flags hallucination by sampling consistency, no evidence	No source sentence to show the reader
Min et al., 2023 (FActScore)	Share of atomic facts a knowledge source supports	A score, not a quote-backed verdict
Ji et al., 2023; Lin et al., 2022	Hallucination survey; TruthfulQA misconceptions	Shows the need, not a tool
Our earlier study	Density signal: AUROC 0.546 on FEVER, 0.275 on AVeriTeC	Does not track support, hence this work

Earlier density signal did not generalise



Reading the chart

Below 0.5 means the signal points the wrong way. Cross-manifold tests fell to chance.

What we do instead

- Read what each source says about the claim
- Require a verbatim quote for every counted source
- Decide with deterministic rules we can prove things about
- Keep density and entropy as research-mode signals only

Method: six steps from answer to traceable verdict

1

Extract

LLM splits the answer into self-contained claims

2

Retrieve

Wikipedia and OpenAlex, merged and deduplicated

3

Rank

Cosine similarity to the claim; keep the top 6

4

Stance

Per source: supports, refutes, neutral, irrelevant, plus a quote

5

Verdict

Rules R1 to R5 over counted sources

6

Show

Side panel: sources, quotes, page highlights

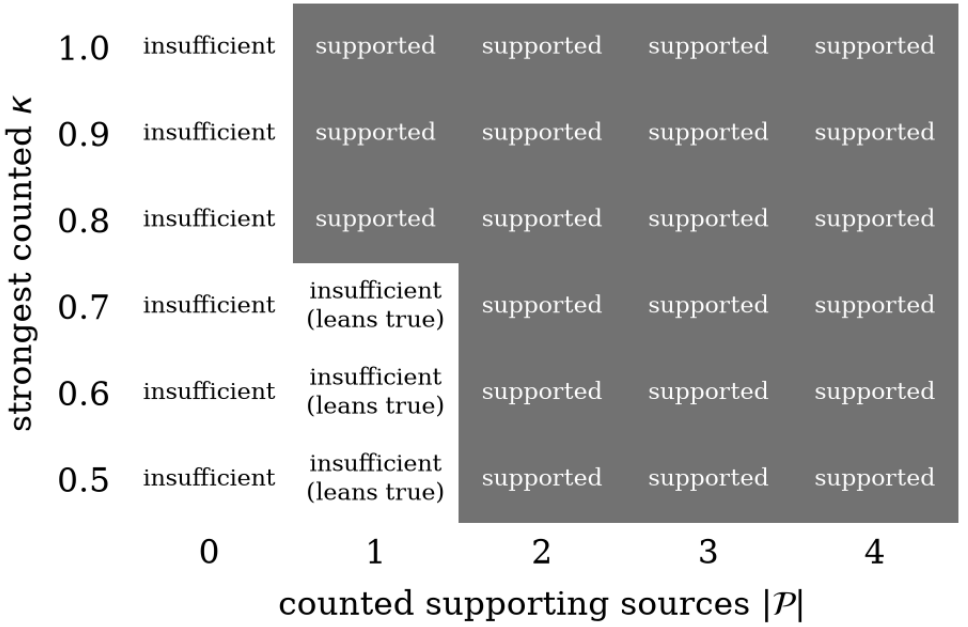
Research mode only
Density ρ (kNN plus Mahalanobis) and entropy U are computed beside the verdict, marked NOT CALIBRATED, and never decide it.

Failure semantics
If every retriever fails, or the stream drops, that claim is an error with a retry. It is never shown as “no sources”.

Verdict rules: five outcomes, decided by counted sources

P: sources that support, confidence ≥ 0.5, quote verified in the source
N: sources that refute, same tests
Q(T): |T| ≥ 2, or |T| = 1 with confidence ≥ 0.8

Rule	Condition (first that applies)	Verdict
R1	no sources retrieved (n = 0)	**no sources**
R2	P and N both non-empty, and Q(P) or Q(N)	**mixed**
R3	Q(P)	**supported**
R4	Q(N)	**refuted**
R5	otherwise	**insufficient**



Verdict by number of counted supporting sources and strongest confidence, no counted refutation. One source qualifies only at confidence 0.8 or more.

Why: one weak source cannot say “backed up”

Statistical and data-science concepts applied

Cosine similarity

Ranks sources by closeness to the claim; keeps the top 6.

kNN + Mahalanobis

Distance of a claim from 10,000 FEVER claim embeddings (research mode).

Ledoit–Wolf shrinkage

Stabilises the covariance; condition number ≤ 1 + (1−δ)p/δ.

Softmax entropy U

Evenness of source matches (τ = 0.1); high when many match alike.

Confusion, macro-F1

Per-class precision and recall; abstention shown as NEI recall.

Wilson 95% interval

Honest error bars for proportions at n = 100 (about ±10 points).

Exact McNemar test

Paired test on discordant claims; every variant sees the same 100 claims.

Paired bootstrap

4,000 resamples for each difference; Bonferroni across six tests.

Five properties, proved and then checked against the code

Theorem 1	**Verdict rules** Exhaustive, exclusive, order-invariant, symmetric, monotone. Every true, false or mixed verdict has a verified quote.
Prop. 2	**Entropy U** Non-decreasing in τ, and $U \geq 1 - \delta/(\tau \ln N)$: many sources matching alike give high U, agree or not.
Prop. 3	**Single-query density** With one query, $\rho \approx (\text{kNN distance} + 1)/2$, so the Mahalanobis term cannot change an AUROC.
Prop. 4	**Shrinkage** Shrunk covariance is positive definite, with condition number $\leq 1 + (1-\delta)p/\delta$.
Prop. 5	**Wilson interval** The Wilson limits form one interval inside [0, 1] that contains the estimate.

27,930

source configurations tested exhaustively for Theorem 1

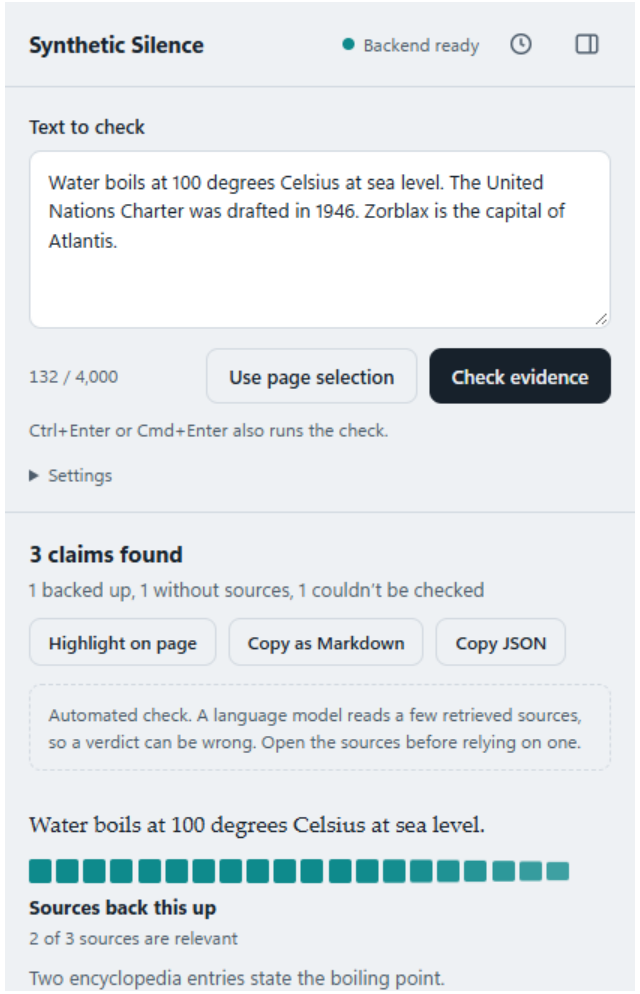
15,000

random configurations of 4 to 7 sources

34

paper tests pass; no counterexample found

Working Chrome extension on a FastAPI backend



Real Chrome 155 run with a stub backend (fixed demo data)

Extension

- Popup and side panel
- Right-click or Alt+Shift+S on any text
- Claims stream in as each one finishes
- Highlight claims on the page, no DOM changes
- History, Markdown and JSON export

Backend

- FastAPI, NDJSON streaming
- Wikipedia and OpenAlex retrieval
- gpt-4o-mini at temperature 0, JSON repair and retry
- Result cache; typed API from OpenAPI
- Budget errors reported, not hidden

117
backend tests pass

44
extension unit tests pass

35
real-Chrome end-to-end checks pass

Confusion matrices on 100 balanced FEVER claims

100	33 / 36 / 31	3	seed 42
claims scored	SUPPORTS / REFUTES / NEI	variants, same claims	gpt-4o-mini, temp 0

Pooled + OpenAlex (baseline)

gold \ pred	S	R	N	M
SUPPORTS	15	9	9	0
REFUTES	3	31	2	0
NEI	2	24	5	0

Pooled + Wikipedia

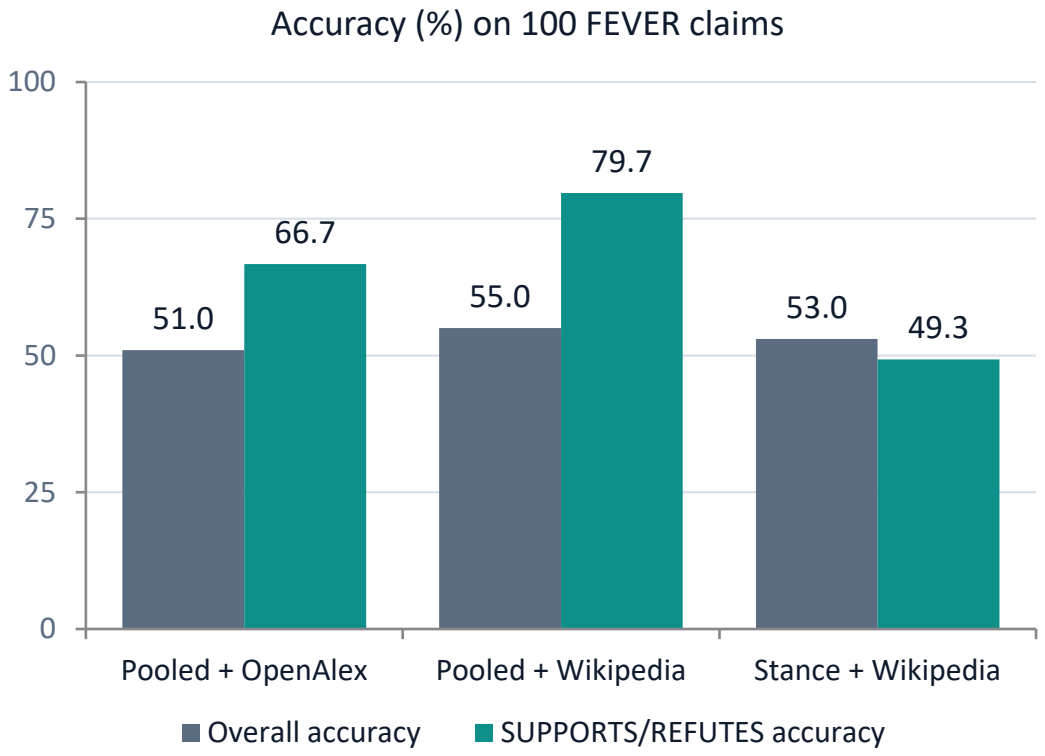
gold \ pred	S	R	N	M
SUPPORTS	24	8	1	0
REFUTES	4	31	1	0
NEI	6	25	0	0

Stance + Wikipedia (first prompt)

gold \ pred	S	R	N	M
SUPPORTS	22	2	8	1
REFUTES	13	12	10	1
NEI	9	3	19	0

Teal: correct cells. Red: wrong cells. Shade is the share of the gold row. S, R, N, M = SUPPORTS, REFUTES, NEI, MIXED. Rows sum to 33, 36, 31.

Results: retrieval helped, pooled never abstained



Variant	Accuracy (95% CI)	Macro-F1	Decisive (95% CI)	NEI recall
Pooled + OpenAlex	51.0 (41.3–60.6)	0.466	66.7 (54.9–76.6)	16.1%
Pooled + Wikipedia	55.0 (45.2–64.4)	0.445	79.7 (68.8–87.5)	0.0%
Stance + Wikipedia	53.0 (43.3–62.5)	0.528	49.3 (37.8–60.8)	61.3%

Percent except macro-F1. Decisive = the 69 SUPPORTS and REFUTES claims.

66.7 → 79.7% decisive accuracy with Wikipedia (suggestive, p = 0.064)

0% vs 61.3% NEI recall: pooled score vs stance variant

Comparison: retrieval mattered more than the rule

Paired tests on the same claims (exact McNemar)

A vs B	Claims	A only / B only	Diff, points (95% CI)	p
PW vs PO	decisive 69	14 / 5	+13.0 (+1.4, +24.6)	0.064
SW vs PW	decisive 69	5 / 26	−30.4 (−44.9, −15.9)	**0.0002**
SW vs PO	decisive 69	11 / 23	−17.4 (−33.3, 0.0)	0.058
SW vs PW	all 100	24 / 26	−2.0 (−16.0, +12.0)	0.888

PO pooled + OpenAlex; PW pooled + Wikipedia; SW stance + Wikipedia. Six tests were run; the Bonferroni threshold is 0.0083, which only the shaded row meets.

The aggregation rule changed decisive accuracy significantly only for the first stance prompt, and for the worse.

Design comparison with related audits

	SelfCheckGPT	FActScore	This work
External evidence	No	Yes	Yes
Unit checked	Sentence	Atomic fact	Claim
Output	Consistency score	Share supported	**Verdict plus quote**

Not compared numerically: the methods use different data and tasks. We claim a difference in design, not in accuracy.

Against our earlier density signal
AUROC 0.546 (FEVER) and 0.275 (AVeriTeC) there; here, verdicts that carry quotes.

Analysis: what we found, and what we got wrong

+13.0 pts

Retrieval source

Wikipedia over OpenAlex on decisive claims. Suggestive (p = 0.064); FEVER is Wikipedia-derived.

0% recall

Pooled score cannot abstain

It called 25 of 31 NEI claims refuted. One score cannot split “not mentioned” from “contradicted”.

p = 0.0002

First stance prompt

Abstains (61.3% NEI recall) but is much worse on decisive claims than the pooled score.

Failure analysis: 13 of 36 REFUTES claims were called supported

- Cause 1: the stance call judged the normalised claim, which had dropped the detail that made it false.
- Cause 2: the prompt accepted near misses (1945 for 1946; a TV series for a radio station).
- **Both fixed with a regression test. The corrected variant is not yet shown to beat pooled, so the default stays pooled.**

Conclusion, limits and next steps

Every verdict traces to a quoted sentence, and absence is kept apart from failure.

Limits

- 100 claims: intervals about ±10 points
- FEVER is Wikipedia-derived: compare variants, not absolutes
- Corrected stance prompt not yet evaluated
- Cross-model audit specified, not run (API credit)

Next steps

- Evaluate the corrected prompt on a 150-claim fresh sample
- Run the 40-prompt audit on seven commercial models (about US$1.6)
- Compare verdicts with human judgements
- Fit the conformal calibration interface (not fitted today)

Thank you. Questions?